



arduino /
arduino-cli



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arduino-cli USB CDC enable function requirement #1869

Closed

Assignees



Labels

type: support



yoyojacky opened on Sep 8, 2022



Describe the request

I want to enable or disable ESP32-C3 module which is from DFRobot, and it works well under arduino-cli, but once i use `arduino-cli monitor -p /dev/ttyACM0` there is no output, after searching and discussing with my friends , he said we need to enable USB CDC during flashing the chip, but i can not find the way to add parameter's about enabling USB CDC.

Describe the current behavior

It is works well under arduino-cli, but once i use `arduino-cli monitor -p /dev/ttyACM0` there is no output, after searching and discussing with my friends , he said we need to enable USB CDC during flashing the chip, but i can not find the way to add parameter's about enabling USB CDC.

Arduino CLI version

0.26.0-> 0.27.1

Operating system

Linux

Operating system version

Debian 11 Bullseye 64bit on Raspberry Pi 4B

Additional context

No response

Closed

arduino-cli USB CDC enable function requirement #1869



I verified the feature was still missing when using the [nightly build](#)

My request contains all necessary details



 **yoyojacky** added `type: enhancement` on Sep 8, 2022



 **per1234** self-assigned this on Sep 8, 2022



per1234 on Sep 8, 2022

Contributor



Hi [@yoyojacky](#). Custom board options such as "**USB CDC On Boot**" are set via the FQBN you pass to `arduino-cli` commands via the `--fqbn` flag.

The format of the FQBN is like this:

```
<vendor ID>:<architecture>:<board ID>[:<menu ID>=<option ID>[,<menu ID>=<option ID>]...]
```



You can learn all the available menu IDs and option IDs for a given board by running [the following command](#):

```
arduino-cli board details --fqbn <FQBN>
```

(where `<FQBN>` is replaced by the fully qualified board name of the board you are using)

For example if you are using the "**ESP32C3 Dev Module**" board (FQBN: `esp32:esp32:esp32c3`):

```
$ ./arduino-cli board details -b esp32:esp32:esp32c3
Board name:      ESP32C3 Dev Module
FQBN:            esp32:esp32:esp32c3
Board version:   2.0.4

Package name:     esp32
Package maintainer: Espressif Systems
Package URL:      https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
Package website:  https://github.com/espressif/arduino-esp32
Package online help: http://esp32.com

Platform name:    esp32
Platform category: ESP32
Platform architecture: esp32
Platform URL:      https://github.com/espressif/arduino-esp32/releases/download/2.0.4/esp32-2.0.4.zip
Platform file name: esp32-2.0.4.zip
```



Platform size (bytes): 259715595
Platform checksum: SHA-
256:832609d6f4cd0edf4e471f02e30b7f0e1c86fdd1b950990ef40431e656237214

Required tool: esp32:riscv32-esp-elf-gcc
gcc8_4_0-esp-2021r2-patch3

Required tool: esp32:xtensa-esp32-elf-gcc
gcc8_4_0-esp-2021r2-patch3

Required tool: esp32:xtensa-esp32s2-elf-gcc
gcc8_4_0-esp-2021r2-patch3

Required tool: esp32:xtensa-esp32s3-elf-gcc
gcc8_4_0-esp-2021r2-patch3

Required tool: esp32:esptool_py
3.3.0

Required tool: esp32:mkspiffs
0.2.3

Required tool: esp32:mklittlefs
3.0.0-gnu12-dc7f933

Option: USB CDC On Boot
CDCOnBoot
Disabled
✓ CDCOnBoot=default
Enabled

CDCOnBoot=cdc
Option: Partition Scheme
PartitionScheme
Default 4MB with spiffs (1.2MB APP/1.5MB SPIFFS)
✓ PartitionScheme=default
Default 4MB with ffat (1.2MB APP/1.5MB FATFS)
PartitionScheme=defaultffat
8M Flash (3MB APP/1.5MB FAT)
PartitionScheme=default_8MB
Minimal (1.3MB APP/700KB SPIFFS)
PartitionScheme=minimal
No OTA (2MB APP/2MB SPIFFS)
PartitionScheme=no_ota
No OTA (1MB APP/3MB SPIFFS)
PartitionScheme=noota_3g
No OTA (2MB APP/2MB FATFS)
PartitionScheme=noota_ffat
No OTA (1MB APP/3MB FATFS)
PartitionScheme=noota_3gffat
Huge APP (3MB No OTA/1MB SPIFFS)
PartitionScheme=huge_app
Minimal SPIFFS (1.9MB APP with OTA/190KB SPIFFS)
PartitionScheme=min_spiffs
16M Flash (2MB APP/12.5MB FAT)
PartitionScheme=fatflash
16M Flash (3MB APP/9MB FATFS)
PartitionScheme=app3M_fat9M_16MB
RainMaker
PartitionScheme=rainmaker
Option: CPU Frequency

CPUFreq	160MHz (WiFi)
✓	CPUFreq=160
	80MHz (WiFi)
CPUFreq=80	
	40MHz
CPUFreq=40	
	20MHz
CPUFreq=20	
	10MHz
CPUFreq=10	
Option:	Flash Mode
FlashMode	
	QIO
✓	FlashMode=qio
	DIO
FlashMode=dio	
	QOUT
FlashMode=qout	
	DOUT
FlashMode=dout	
Option:	Flash Frequency
FlashFreq	
	80MHz
✓	FlashFreq=80
	40MHz
FlashFreq=40	
Option:	Flash Size
FlashSize	
	4MB (32Mb)
✓	FlashSize=4M
	8MB (64Mb)
FlashSize=8M	
	2MB (16Mb)
FlashSize=2M	
	16MB (128Mb)
FlashSize=16M	
Option:	Upload Speed
UploadSpeed	
	921600
✓	UploadSpeed=921600
	115200
UploadSpeed=115200	
	256000
UploadSpeed=256000	
	230400
UploadSpeed=230400	
	512000
UploadSpeed=512000	
Option:	Core Debug Level
DebugLevel	
	None
✓	DebugLevel=none
	Error
DebugLevel=error	
	Warn
DebugLevel=warn	
	Info
DebugLevel=info	
	Debug

```
DebugLevel=debug
Verbose
DebugLevel=verbose
Programmers:
Id
Name
esptool
Esptool
```

Here we can see that the ID for the **"USB CDC On Boot"** menu is `CDC0nBoot` and the ID for the **"Enabled"** option is `cdc`.

So the full FQBN is:

```
esp32:esp32:esp32c3:CDC0nBoot=cdc
```

If you have any additional questions about using Arduino CLI, feel free to post on the dedicated Arduino Forum section here:

<https://forum.arduino.cc/c/software/arduino-cli/89>

More information about custom board options here:

<https://arduino.github.io/arduino-cli/latest/platform-specification/#custom-board-options>

And about setting them via the FQBN here:

<https://arduino.github.io/arduino-cli/latest/FAQ/#how-to-set-multiple-board-options>

We are tracking the need to better document the FQBN format here:

[#355](#)



 **per1234** closed this as not planned on Sep 8, 2022

  **per1234** added `type: support` and removed `type: enhancement` on Sep 8, 2022

 **yoyojacky** on Sep 8, 2022

Author ...

Hi [@yoyojacky](#). Custom board options such as **"USB CDC On Boot"** are set via the FQBN you pass to `arduino-cli` commands via the `--fqbn` flag.

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```



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```
arduino-cli board details --fqbn <FQBN>
```



(where <FQBN> is replaced by the fully qualified board name of the board you are using)

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```
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Board name:      ESP32C3 Dev Module
FQBN:            esp32:esp32:esp32c3
Board version:   2.0.4

Package name:     esp32
Package maintainer: Espressif Systems
Package URL:      https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
Package website:  https://github.com/espressif/arduino-esp32
Package online help: http://esp32.com

Platform name:     esp32
Platform category: ESP32
Platform architecture: esp32
Platform URL:      https://github.com/espressif/arduino-esp32/releases/download/2.0.4/esp32-2.0.4.zip
Platform file name: esp32-2.0.4.zip
Platform size (bytes): 259715595
Platform checksum: SHA-
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Required tool:     esp32:riscv32-esp-elf-gcc
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Required tool:     esp32:xtensa-esp32-elf-gcc
gcc8_4_0-esp-2021r2-patch3

Required tool:     esp32:xtensa-esp32s2-elf-gcc
gcc8_4_0-esp-2021r2-patch3

Required tool:     esp32:xtensa-esp32s3-elf-gcc
gcc8_4_0-esp-2021r2-patch3

Required tool:     esp32:esptool_py
3.3.0

Required tool:     esp32:mkspiffs
0.2.3

Required tool:     esp32:mklittlefs
3.0.0-gnu12-dc7f933

Option:            USB CDC On Boot
CDCOnBoot
Disabled
✓                  CDCOnBoot=default
Enabled
```



```

CDCOnBoot=cdc
Option:          Partition Scheme
PartitionScheme

                Default 4MB with spiffs (1.2MB APP/1.5MB SPIFFS)
✓                PartitionScheme=default
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PartitionScheme=defaultffat
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PartitionScheme=default_8MB
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PartitionScheme=minimal
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PartitionScheme=no_ota
                No OTA (1MB APP/3MB SPIFFS)
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PartitionScheme=noota_ffat
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PartitionScheme=fatflash
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PartitionScheme=app3M_fat9M_16MB
                RainMaker
PartitionScheme=rainmaker
Option:          CPU Frequency
CPUFreq

                160MHz (WiFi)
✓                CPUFreq=160
                80MHz (WiFi)
CPUFreq=80
                40MHz
CPUFreq=40
                20MHz
CPUFreq=20
                10MHz
CPUFreq=10
Option:          Flash Mode
FlashMode

                QIO
✓                FlashMode=qio
                DIO
FlashMode=dio
                QOUT
FlashMode=qout
                DOUT
FlashMode=dout
Option:          Flash Frequency
FlashFreq

                80MHz
✓                FlashFreq=80
                40MHz
FlashFreq=40
Option:          Flash Size
FlashSize

                4MB (32Mb)

```

✓	FlashSize=4M
	8MB (64Mb)
FlashSize=8M	
	2MB (16Mb)
FlashSize=2M	
	16MB (128Mb)
FlashSize=16M	
Option:	Upload Speed
UploadSpeed	
	921600
✓	UploadSpeed=921600
	115200
UploadSpeed=115200	
	256000
UploadSpeed=256000	
	230400
UploadSpeed=230400	
	512000
UploadSpeed=512000	
Option:	Core Debug Level
DebugLevel	
	None
✓	DebugLevel=none
	Error
DebugLevel=error	
	Warn
DebugLevel=warn	
	Info
DebugLevel=info	
	Debug
DebugLevel=debug	
	Verbose
DebugLevel=verbose	
Programmers:	Id
Name	
	esptool
Esptool	

Here we can see that the ID for the **"USB CDC On Boot"** menu is `CDC0nBoot` and the ID for the **"Enabled"** option is `cdc`.

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Thank you very much!!! It works!!! you are my god.... thank you again.



 **yoyojacky** closed this as completed on Sep 8, 2022

  **yoyojacky** mentioned this on Sep 8, 2022

 Native USB CDC hangs up on ESP32-S2 after about 320 characters [espressif/arduino-esp32#6221](#)



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